Research Update

14 October 2026

Project 1: Sharing or Specializing? A Systematic Study of Parameter-Efficient Vision–Language Model Adaptation Across Heterogeneous Medical Tasks (IEEE TMI)

When adapting a VLM to heterogeneous medical imaging tasks, which adaptation parameters should be shared and which should be specialized?

1.1 Research questions

- **RQ1.** Does task-specific adaptation outperform shared adaptation, and for which task families?
- **RQ2.** How does the amount of task-specific supervision change the optimal degree of sharing?
- **RQ3.** What determines whether tasks help or hurt each other: output format, imaging modality, or dataset?
- **RQ4.** Which measurable task characteristics predict whether tasks benefit from shared or specialized adaptation?

1.2 Design

- **Backbones:** Qwen3.5-9B and MedGemma.
- **Data:** FLARE 2026 Task 3 (17 datasets, 7 modalities, 6 task families).

Degree of sharing

Strategy	A matrix (input projection)	B matrix (output projection)	Degree of sharing	Parameters per layer	What it tells us
Shared LoRA	Shared	Shared	Full	$1A + 1B$	Can one adapter serve all tasks?
HydraLoRA (task-assigned heads)	Shared	Task-specific	Partial	$1A + 6B$	Do tasks share input features but need their own output mapping?
Reversed HydraLoRA	Task-specific	Shared	Partial	$6A + 1B$	Does it matter which matrix is shared, or only how much?
Task LoRA	Task-specific	Task-specific	None	$6A + 6B$	Is full specialization worth six times the parameters?
Shared LoRA, rank $\times$ 6 (control)	Shared	Shared	Full	$\approx 6A + 6B$	Is a task-LoRA gain due to specialization or only to more parameters?